

**ILOILO STATE UNIVERSITY OF FISHERIES SCIENCE AND
TECHNOLOGY**

**COLLEGE OF INFORMATICS AND COMPUTING
INNOVATIONS**

Software Test Description
for
Event Management System

Change History

Version	Date	Description of Changes	Author
1.0	30 August 2026	Initial Software Test Description	Project Team
1.1	02 September 2026	Updated test cases/procedures	Test Lead
Final	13 September 2026	Approved final version	Project Team

Preface

This Software Test Description (STD) defines the planned testing activities for the School Event Management System. It translates the system requirements and design into test objectives, test cases, test data, procedures, expected results, and acceptance criteria.

This document is prepared as a project testing artifact and should be updated whenever approved system requirements, design, or implementation changes affect the test scope.

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1. Introduction

1.1. Purpose

The purpose of this Software Test Description is to define how the School Event Management System will be tested to determine whether its implemented functions satisfy the approved requirements and operate correctly. Testing will provide evidence for verification, validation, defect correction, and final user acceptance.

1.2. Scope

Testing covers the core functions identified for the School Event Management System: authentication, role-based access, event creation and management, event approval, event registration, announcements, attendance monitoring, search/filtering, and reports.

Testing will include functional, integration, system, usability, security-oriented access-control checks, and user acceptance testing. Financial accounting, academic records, and unrelated university services are outside the test scope.

1.3. Definitions, Acronyms and Abbreviations

STD – Software Test Description

SEMS – School Event Management System

SRS – Software Requirements Specification

SDD – Software Design Description

UAT – User Acceptance Testing

QA – Quality Assurance

UI – User Interface

DB – Database

TC – Test Case

FR – Functional Requirement

2. References

The test scope and terminology are based on the School Event Management System SRS and SDD prepared for this project. The supplied SDD template organizes system design

into modules, processes, data, interfaces, detailed design, and annexed diagrams.

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The supplied SPMP template also identifies component, integration, system, and acceptance testing as project verification and validation activities.

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The project SRS defines the intended functional scope of authentication, event management, approval, registration, announcements/notifications, attendance, search/filtering, and reporting.

3. Test Strategy

3.1. Testing Objectives

- Verify that each implemented function conforms to its approved requirement.
- Detect defects before system acceptance.
- Verify data is created, updated, retrieved, and related correctly.
- Verify role-based access prevents unauthorized functions.
- Verify event registration handles capacity and duplicate-registration rules.
- Verify attendance and reports reflect stored data accurately.
- Provide evidence for stakeholder acceptance.

3.2. Testing Levels

Level	Purpose	Primary Responsibility
Component/Unit Testing	Verify individual modules or functions.	Developer
Integration Testing	Verify interactions between modules and database/services.	Developer + Tester
System Testing	Verify the complete integrated system against requirements.	Tester/QA
Acceptance Testing	Determine whether the system meets stakeholder	Stakeholder + Tester

acceptance criteria.

3.3. Testing Approach

Testing will use requirement-based functional tests supported by positive, negative, boundary, role-based, data-integrity, usability, and regression checks. Test cases will be mapped to system functions. Failed tests will produce defect records and may be re-executed after correction.

3.4. Test Environment

Environment Item	Planned Configuration
Client devices	Desktop/laptop and mobile device where available
Browsers	Google Chrome, Microsoft Edge, Mozilla Firefox or supported equivalents
Application server	Development/test server or approved local environment
Database	MySQL or selected relational database
Network	Reliable local network/internet connection
Test repository	Version-controlled project repository and test records

3.5. Test Data

- Valid administrator account
- Valid organizer account
- Valid student/participant account
- Invalid and incomplete login credentials
- Sample approved and pending events
- Events at, below, and beyond capacity
- Duplicate registration attempts
- Sample attendance records
- Sample announcements

- Completed/cancelled events
- Sample report data

3.6. Entry and Exit Criteria

Entry criteria: a testable build is available; relevant requirements/design are approved; test environment is configured; test data is prepared; and critical dependencies are operational.

Exit criteria: planned test cases have been executed; critical/high defects are resolved or formally accepted; required regression tests pass; test results are documented; and acceptance testing is completed or formally waived.

4. Test Items

4.1. User Authentication

Login, logout, credential validation, session protection, and unauthorized-access handling.

4.2. User and Role Management

Account creation/update/deactivation and role-based permissions.

4.3. Event Management

Create, edit, save, publish, update, cancel, and view event records.

4.4. Event Approval

Submit, review, approve/reject, and track event proposal status.

4.5. Event Registration

Open registration, eligibility, capacity, duplicate prevention, and registration status.

4.6. Announcements and Notifications

Create and display event announcements and relevant updates.

4.7. Attendance Monitoring

Record and view participant attendance.

4.8. Search and Filtering

Search events and filter by available event attributes.

4.9. Reports

Generate participant, attendance, and event summary reports.

5. Test Cases

5.1. Authentication Test Cases

ID	Test	Input/Precondition	Expected Result	Status
TC-AUTH-01	Valid login	Valid user credentials	Dashboard opens according to role; authentication succeeds.	Pass/Fail
TC-AUTH-02	Invalid password	Valid username + wrong password	Login is rejected and error message is shown.	Pass/Fail
TC-AUTH-03	Empty credentials	Blank required fields	Validation prevents submission or reports required fields.	Pass/Fail
TC-AUTH-04	Logout	Authenticated user	Session ends and protected pages require authentication.	Pass/Fail
TC-AUTH-05	Role restriction	Participant account accesses admin URL/function	Access is denied.	Pass/Fail

5.2. Event Management Test Cases

ID	Test	Input/Precondition	Expected Result	Status
TC-EVT-01	Create event	Organizer enters all required details	Event is saved as draft/pending according to	Pass/Fail

			workflow.	
TC-EVT-02	Required-field validation	Missing title/date/venue	System identifies missing required data.	Pass/Fail
TC-EVT-03	Edit event	Existing organizer-owned event	Permitted changes are saved correctly.	Pass/Fail
TC-EVT-04	Approve event	Administrator reviews pending event	Status changes to Approved and event can be published.	Pass/Fail
TC-EVT-05	Reject event	Administrator rejects pending event	Status changes to Rejected and result is recorded.	Pass/Fail
TC-EVT-06	Cancel event	Authorized user cancels event	Status displays Cancelled and participant-facing information is updated.	Pass/Fail

5.3. Registration Test Cases

ID	Test	Input/Precondition	Expected Result	Status
TC-REG-01	Register for open event	Eligible participant; event has capacity	Registration succeeds and status is displayed.	Pass/Fail
TC-REG-02	Duplicate registration	Same participant registers twice	Second registration is rejected/prevented.	Pass/Fail
TC-REG-03	Full event	Event capacity reached	New registration is prevented.	Pass/Fail

TC-REG-04	Closed registration	Registration period ended	Registration action is unavailable or rejected.	Pass/Fail
TC-REG-05	Participant list	Authorized organizer views event	Registered participants are displayed accurately.	Pass/Fail

5.4. Attendance Test Cases

ID	Test	Input/Precondition	Expected Result	Status
TC-ATT-01	Record attendance	Registered participant selected	Attendance record is saved.	Pass/Fail
TC-ATT-02	Update attendance	Existing attendance record	Authorized change is saved correctly.	Pass/Fail
TC-ATT-03	Unauthorized attendance access	Participant attempts organizer attendance function	Access is denied.	Pass/Fail
TC-ATT-04	Attendance summary	Event with attendance records	Summary matches stored attendance data.	Pass/Fail

5.5. Reporting Test Cases

ID	Test	Input/Precondition	Expected Result	Status
TC-RPT-01	Participant report	Event with registrations	Participant list/report is generated correctly.	Pass/Fail
TC-RPT-02	Attendance	Completed event	Attendance	Pass/Fail

	report	with attendance	report reflects recorded data.	
TC-RPT-03	Event summary	Multiple events with statuses	Summary displays correct event information.	Pass/Fail

6. Test Procedures

6.1. Functional Test Procedure

- Select the approved build and test case.
- Prepare the required test account and data.
- Execute the specified user actions.
- Record actual results.
- Compare actual result with expected result.
- Mark Pass or Fail.
- For failures, create a defect record with evidence.
- Re-test after correction.

6.2. Integration Test Procedure

Test connected workflows such as login → role access; event creation → approval → publication; event publication → registration; registration → attendance; and stored data → reporting. Confirm that data passed between modules remains complete, accurate, and consistent.

6.3. System Test Procedure

Execute the complete set of prioritized functional and non-functional checks using a production-like test environment. Include normal, invalid, boundary, role-based, and regression scenarios.

6.4. User Acceptance Test Procedure

Selected university stakeholders will execute representative real-world workflows. Feedback will be recorded. The system may be accepted when agreed acceptance criteria are satisfied and no unresolved critical defects remain.

7. Defect Management

7.1. Defect Classification

Severity	Description	Example
Critical	Prevents core system operation, causes severe data/security failure.	Users cannot log in or data is corrupted.
High	Major function fails with significant impact.	Participants cannot register for any event.
Medium	Function works incorrectly but a workaround exists.	One report field displays incorrect formatting.
Low	Minor cosmetic/usability issue.	Small alignment or wording problem.

7.2. Defect Life Cycle

New → Assigned → In Progress → Fixed → Ready for Retest → Verified/Closed. If the defect remains, it returns to Assigned/In Progress. Rejected or duplicate defects are documented with a reason.

8. Test Schedule and Responsibilities

Activity	Target	Responsible	Output
Test planning	Week 10–11	Test Lead	STD/test cases
Component testing	Week 10–11	Developers	Unit/component results
Integration testing	Week 11	Developer + Tester	Integration results
System testing	Week 11–12	Tester/QA	System test report
Defect correction/retest	Week 12	Developers + Tester	Updated results
Acceptance testing	Week 13	Stakeholders + Tester	UAT results
Final test report	Week 14	Test Lead	Approved test documentation

9. Test Deliverables

- Approved Software Test Description.
- Test case list and test data.
- Test execution records.
- Defect/issue log.
- Regression-test results.
- System test report.
- User acceptance test results.
- Final test summary and recommendation.

The project SPMP template recognizes component, integration, system, and acceptance testing and identifies test cases, test procedures, anomaly reports, V&V summaries, and final reports as relevant outputs. ☐filecite☐turn6file6☐L713-L802☐

10. Appendices

Appendix A – Test Case Execution Record

For each test case, record: Test ID, date, tester, build/version, environment, preconditions, input data, actual result, expected result, Pass/Fail, defect ID if applicable, and remarks.

Appendix B – Requirements Traceability

Each approved functional requirement should map to one or more test cases. Example: Authentication requirements → TC-AUTH-01 through TC-AUTH-05; Event Management → TC-EVT-01 through TC-EVT-06; Registration → TC-REG-01 through TC-REG-05; Attendance → TC-ATT-01 through TC-ATT-04; Reports → TC-RPT-01 through TC-RPT-03.

Appendix C – Acceptance Checklist

- Authentication and role access tested
- Event creation and approval tested
- Registration rules tested
- Announcements tested
- Attendance tested

- Search/filter tested
- Reports tested
- Critical/high defects resolved or accepted
- UAT completed
- Final test report approved

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